

Solution Mid Sem (MCC505)

1(a)

$$n_1 = 100 \quad \bar{x}_1 = 15, \sigma_1 = 3$$

$$n_2 = 150 \quad \bar{x}_2 = 25, \sigma_2 = 5$$

$$n_3 = 200 \quad \bar{x}_3 = 30, \sigma_3 = 7$$

$$n_4 = 250 \quad \bar{x}_4 = ?, \sigma_4 = 6$$

$$n = n_1 + n_2 + n_3 + n_4 \\ = 700$$

Combined mean $\bar{x} = 22.5$

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3 + n_4 \bar{x}_4}{n}$$

$$\bar{x}_4 = \left[n\bar{x} - (n_1 \bar{x}_1 + n_2 \bar{x}_2 + n_3 \bar{x}_3) \right] / n_4$$

$$= \left[700 \times 22.5 - (100 \times 15 + 150 \times 25 + 200 \times 30) \right] / 250 \\ = (15750 - (1500 + 3750 + 6000)) / 250 \\ = (15750 - 11250) / 250 = \frac{4500}{250} = 18$$

$$\bar{x}_4 = 18 \quad \text{--- (1 mark)}$$

$$\text{Combined variance } \sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2) + n_3(\sigma_3^2 + d_3^2) + n_4(\sigma_4^2 + d_4^2)}{n}$$

$$d_1 = \bar{x}_1 - \bar{x} = 15 - 22.5 = -7.5$$

$$d_2 = \bar{x}_2 - \bar{x} = 25 - 22.5 = 2.5$$

$$d_3 = \bar{x}_3 - \bar{x} = 30 - 22.5 = 7.5$$

$$d_4 = \bar{x}_4 - \bar{x} = 18 - 22.5 = -4.5$$

$$\sigma^2 = \frac{100(9 + 56.25) + 150(25 + 6.25) + 200(49 + 56.25) + 250(36 + 20.25)}{700}$$

$$= \frac{6525 + 4687.5 + 21050 + 14062.5}{700} = \frac{46325}{700}$$

$$\sigma = 8.14$$

$$CV = \frac{\sigma}{\bar{x}} = \frac{8.14}{22.5} = 0.3618 \quad \text{--- (1 mark)}$$

= 66.18% --- (2 mark)

1(b)

C.I	f	C.F. (LTT)
0-20	30	30
20-30	195	225
30-40	240	465
40-50	115	580
50-60	54	634
60-70	10	644
70-80	6	650
80-90	15	665
90-100	15	680

N=680

Coefficient of Skewness

$$S_K = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$$

Where

$$Q_i = 1 + \frac{h}{f} \left(\frac{iN}{4} - C \right) \quad i=1,2,3$$

- i) For identify Q_1 class, calculate $\frac{N}{4} = \frac{680}{4} = 170$, hence
20-30 is the Q_1 class

$$\therefore Q_1 = 1 + \frac{h}{f} \left(\frac{N}{4} - C \right) = 20 + \frac{10}{195} (170 - 30) = 27.18$$

- ii) For identify Q_2 class, calculate $\frac{2N}{4} = \frac{N}{2} = \frac{680}{2} = 340$, hence
30-40 is the Q_2 class

$$\therefore Q_2 = 1 + \frac{h}{f} \left(\frac{N}{2} - C \right) = 30 + \frac{10}{240} (340 - 225) = 34.79$$

- iii) For identify Q_3 class, calculate $\frac{3N}{4} = \frac{3 \times 680}{4} = 510$, hence
40-50 is the Q_3 class

$$\therefore Q_3 = 1 + \frac{h}{f} \left(\frac{3N}{4} - C \right) = 40 + \frac{10}{115} (510 - 465) = 43.91$$

Hence, $Q_1 = 27.18$, $Q_2 = 34.79$, $Q_3 = 43.91$

$$\therefore S_K = \frac{(43.91 - 34.79) - (34.79 - 27.18)}{(43.91 - 34.79) + (34.79 - 27.18)} = \frac{9.12 - 7.61}{9.12 + 7.61} = \frac{1.51}{16.73} = 0.0903$$

$S_K = 0.0903$

Marking Pattern

- Q_1 — (1 mark)
 Q_2 — (1 mark)
 Q_3 — (1 mark)
 S_K — (1 mark)